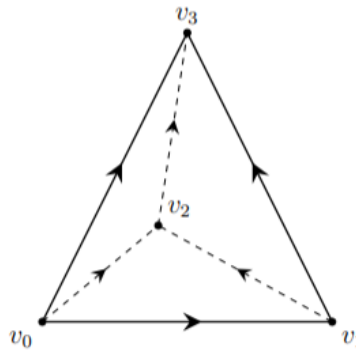


MTH 732 Topology - Homework 7

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Assignment 1. Consider the following 3-simplex, which we denote by $[v_0, v_1, v_2, v_3]$.



We compute its boundary and label its faces as follows:

$$\partial[v_0, v_1, v_2, v_3] = \underbrace{[v_1, v_2, v_3]}_A - \underbrace{[v_0, v_2, v_3]}_B + \underbrace{[v_0, v_1, v_3]}_C - \underbrace{[v_0, v_1, v_2]}_D.$$

Let X be the space resulting from gluing faces A to B and C to D . Compute the simplicial homology groups $H_n^\Delta(X)$. *Hint: there should be one 3-simplex, two 2-simplices, three 1-simplices, and two 0-simplices.* What space is this?

Proof. Here's my attempt at a picture:



- The 0-simplices are $[v_1]$ and $[v_2]$. Thus $\ker \partial_0 = \mathbb{Z} \langle [v_1], [v_2] \rangle$.

- The 1-simplices are $[v_1, v_1]$, $[v_1, v_2]$, and $[v_2, v_2]$. Thus $\ker \partial_1 = \mathbb{Z} \langle [v_1, v_1], [v_2, v_2] \rangle$ and $\text{im } \partial_1 = \mathbb{Z} \langle [v_2 - v_1] \rangle$.
- The 2-simplices are $[v_1, v_1, v_1]$ and $[v_1, v_2, v_2]$. Thus $\partial_2[v_1, v_1, v_1] = [v_1, v_1]$, and $\partial_2[v_1, v_2, v_2] = [v_2, v_2]$, so $\ker \partial_2 = 0$ and $\text{im } \partial_2 = \mathbb{Z} \langle [v_1, v_1], [v_2, v_2] \rangle$.
- The 3-simplex is $[v_1, v_1, v_2, v_2]$, so $\partial_3 = 0$.

Since there are no 4-simplices whose boundaries we can mod out by, we immediately get $H_3^\Delta(X) \cong \mathbb{Z}$. We also immediately get $H_2^\Delta(X) \cong 0$ since $\ker \partial_2 = 0$. We now get $H_1^\Delta(X) = \mathbb{Z} \langle [v_1, v_1], [v_2, v_2] \rangle / \mathbb{Z} \langle [v_1, v_1], [v_2, v_2] \rangle \cong 0$. Finally, we have $H_0^\Delta(X) = \mathbb{Z} \langle [v_1], [v_2] \rangle / \mathbb{Z} \langle [v_2 - v_1] \rangle \cong \mathbb{Z}$.

My guess for what this space was was $S^1 \vee S^2 \vee S^2$. Visually, that's what it looks like, but the homology doesn't match up, so I'm not sure what I would call this. $\square$

Assignment 2. Are there group homomorphisms f and g such that

$$0 \rightarrow \mathbb{Z}_4 \xrightarrow{f} \mathbb{Z}_8 \oplus \mathbb{Z}_2 \xrightarrow{g} \mathbb{Z}_4 \rightarrow 0$$

is a short exact sequence?

Proof. Define

$$\begin{aligned} f : 1 &\mapsto (2, 1) \\ g : (x, y) &\mapsto x + 2y. \end{aligned}$$

Then f is injective as required since

$$\begin{aligned} f(1) &= (2, 1) \\ f(2) &= (4, 0) \\ f(3) &= (6, 1) \\ f(0) &= (0, 0) \end{aligned}$$

and g is surjective as required since

$$\begin{aligned} g(0, 0) &= 0 \\ g(1, 0) &= 1 \\ g(0, 1) &= 2 \\ g(1, 1) &= 3. \end{aligned}$$

The other requirement for exactness is that $\text{im } f = \ker g$. Indeed, on the image of f as listed above, $g = 0$, and if (x, y) were any other element of $\mathbb{Z}_8 \oplus \mathbb{Z}_2$, we would have x being “odd”, and since $2y$ is always “even”, we could not end up with $x + 2y = 0$, so $\ker g = \text{im } f$, and the sequence is exact. $\square$

Assignment 3. Let

$$A \xrightarrow{f} B \rightarrow C \rightarrow D \xrightarrow{g} E$$

be an exact sequence. Show that $C = 0$ (trivial group) if and only if f is surjective **and** g is injective.

Proof. We put a couple extra labels on the diagram like so:

$$A \xrightarrow{f} B \xrightarrow{\phi} C \xrightarrow{\psi} D \xrightarrow{g} E .$$

If $C = 0$, then $\text{im } f = \ker \phi = B$ and $\ker g = \text{im } \psi = 0$. Thus f is surjective and g is injective as desired.

If we first have f being surjective and g being injective, then we know that $\ker \phi = \text{im } f = B$ and $\text{im } \psi = \ker g = 0$. This forces f to be the trivial map since B is the kernel, so $\ker \psi = \text{im } \phi = 0$, meaning ψ is injective. Since $\text{im } \psi = 0$, we conclude that C has only one element and is thus the trivial group, as desired. $\square$